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Shotcrete for Dependencies

Secure package management with NuGet

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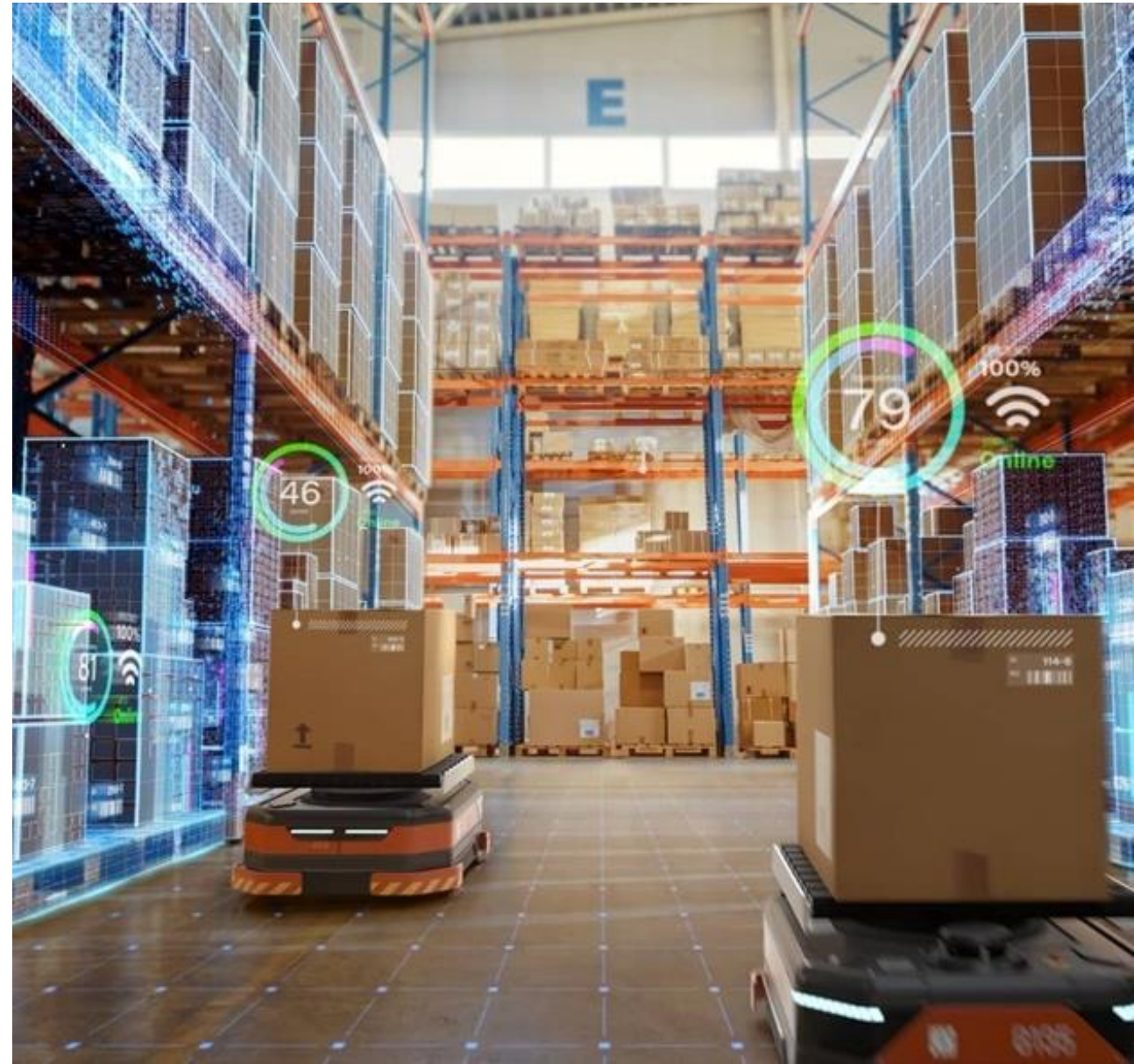
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Agenda

1. bbv in brief
2. Why dependency management matters
3. Central package management
4. NuGet Audit
5. Lock files
6. Bonus
7. Q&A





bbv in brief

Facts, figures & interesting things to know

Our vision and principles



The way we work reflects a set of principles which we strive to practise in the interests of our clients and partners. This is how we would like them to perceive us:



excellent



customer focussed



co-responsible



respectful



passionate

bbv Software Services is a Swiss software and consulting company with the aim of contributing its expertise to the most important visions, projects and challenges for mankind in our interconnected and globalised world.

Facts & figures

30 years bbv



300 employees

200 in Switzerland
100 abroad



8 offices

Switzerland, Germany,
Greece, Vietnam



38,2 million Swiss francs

Sales 2024





Digital Business

- Digital Business Models
- B2B e-commerce platform
- Product Vision
- New Revenue Streams
- Cybersecurity
- Organisational Design



Digital Experience

- Product & Service Design
- User & Customer Experience
- Digital Marketing



Products & Services

- **Software Engineering**
- **Software Architecture**
- **Cloud Services**
- **Modernisation**
- **Mobile & Web Apps**
- **DevOps & ALM**
- Generative AI
- Data & Analytics
- Managed Teams
- bbv AI Hub
- Internet of Things
- Shoring
- Swiss LCDM Hub
- BIM2FM
- Agentic Pimcore

bbv Academy – Trends in a nutshell

Benefit from our experts' knowhow in established and emerging technologies and methods:

- Over 40 practical training programmes
- Proven mix of theory and hands-on
- Flexible venues

Seminars



Workshops



Certifications



Customised training

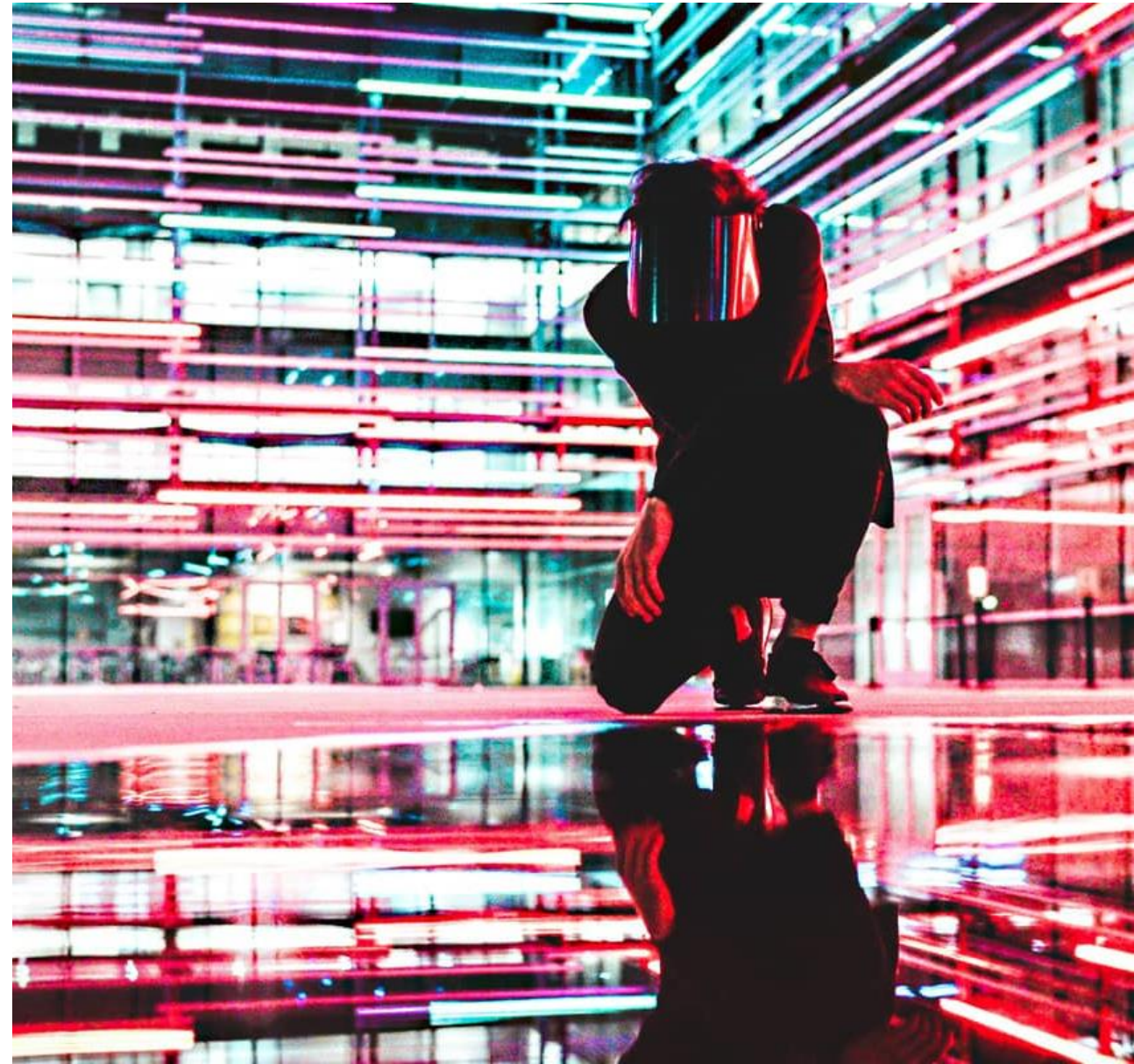


Visit the bbv Academy 

Why dependency management matters

1. Security
2. Compliance
3. DevOps

supply chain attacks, dependency confusion, typosquatting package changes (updates, deletions), transitive dependencies, SBOM, license management, visible changes, inconsistent `nuget.configs`, targets, analyzers, source generators, tools



Central package management

```
<PropertyGroup>
```

```
  <ManagePackageVersionsCentrally>
```

```
    ...
```

```
  <CentralPackage  
    TransitivePinningEnabled>
```

```
    ...
```

```
  <CentralPackage  
    VersionOverrideEnabled>
```

```
    ...
```

```
</PropertyGroup>
```



```
<GlobalPackageReference ...
```

Asset management



NuGet Audit

Find (known) security issues in dependencies during development.

```
2021: dotnet list package --vulnerable
```

...

```
2024 (opt-in): dotnet restore
```

```
2025 (default): dotnet restore
```

```
<PropertyGroup>
```

```
  <NuGetAuditMode>all
```

```
<ItemGroup>
```

```
  <NuGetAuditSuppress Include=  
    "https://github.com/advisories/XXXX"
```



Lock files

<RestorePackagesWithLockFile>true

```
"Asp.Versioning.Abstractions": {  
  "type": "Transitive",  
  "resolved": "8.1.0",  
  "contentHash": "mpeNZyMdvRHztJwR1sXIUQ+3iioEU97YMBnFA9WLbsPOYhGwDjNqJmEd8ny7kcmS90jThEuX/bSXXY3brIFA=",  
  "dependencies": {  
    "Microsoft.Extensions.Primitives": "8.0.0"  
  }  
},  
"Asp.Versioning.Http": {  
  "type": "Transitive",  
  "resolved": "8.1.0",  
  "contentHash": "Xu4xF62Cu9JqYi/CTa2TiK5kyHoa4ELuPynj/bPFWDmLTIPzuJQbBI5RgFYVRFHjFVvWmoA77acRaFu7i7Wzqg=",  
  "dependencies": {  
    "Asp.Versioning.Abstractions": "8.1.0"  
  }  
},  
"Asp.Versioning.Mvc": {  
  "type": "Transitive",  
  "resolved": "8.1.0",  
  "contentHash": "BMAJM2sGsTUw5FQ9upKQt6GFoLdWksePgGpYj156WSRvIuE3UxKZh0gAL+wDTIfLshUZm97VCVx10GyrcjWz9Q=",  
  "dependencies": {  
    "Asp.Versioning.Http": "8.1.0"  
  }  
},  
}
```

dotnet restore --locked-mode

dotnet build --no-restore

- ⚠ Multi architecture and target builds
- ⚠ new resolution strategy (pre .NET 10)
- ⚠ dependabot, renovate



Bonus (a.k.a nuget.config)

```
<?xml version="1.0" encoding="utf-8"?>
<configuration>
  <packageSources>
    <clear />
    <add key="nuget" value="..." />
    <add key="myCorp" value="..." />
  </packageSources>
  <auditSources>
    <clear />
    <add key="nuget" value="..." />
  </auditSources>
  <packageSourceMapping>
    <clear />
    <packageSource key="nuget.org">
      <package pattern="*" />
    </packageSource>
    <packageSource key="myCorp">
      <package pattern="My.Corp.*" />
    </packageSource>
  </packageSourceMapping>
  <trustedSigners>
    ...
  </trustedSigners>
</configuration>
```





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